



Statistics of Financial Markets

Chapter 4: Probability



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Acronyms used in this material

- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- ARCH: AutoRegressive Conditional Heteroskedasticity
- BET: Bucharest Exchange Trading (index)
- CDF: Cumulative Distribution Function
- CLT: Central Limit Theorem
- CRR: Cox–Ross–Rubinstein (binomial model)
- ES: Expected Shortfall
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GBM: Geometric Brownian Motion
- IBM: International Business Machines (company)
- LCG: Linear Congruential Generator
- LLN: Law of Large Numbers
- PCG64: Permuted Congruential Generator, 64-bit (NumPy default)
- PMF: Probability Mass Function
- RANDU: RANDU (IBM random number generator of the 1960s)
- SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

Today's Question and Route

- ▣ **Question:** which probability tools do we need to describe returns and to simulate prices?
 - ▶ every later chapter (tails, volatility, VaR, efficiency) is built on them
- ▣ **Route** of the chapter
 - ▶ random variables, distributions, expectation, variance, covariance
 - ▶ independence vs uncorrelatedness, conditional expectation and conditional variance
 - ▶ random numbers and Monte Carlo; the binomial model
 - ▶ random walk, martingale, white noise, AR(1); Wiener process and GBM
- ▣ Real data, 2000–2026: S&P 500, DAX, BET and Bitcoin (from 2014)

Learning Outcomes

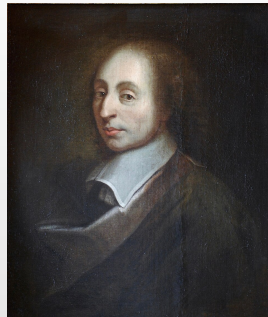
- Compute expectations, variances, covariances and correlations, and the volatility of a portfolio
- Explain why uncorrelated returns can still be dependent, and measure it
- Use conditional expectation and conditional variance, the building blocks of GARCH
- Generate random numbers with the inverse transform and estimate a probability by Monte Carlo, with its standard error
- Define a random walk, a martingale, white noise, the Wiener process and GBM, and simulate them

Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), *Statistics of Financial Markets*, 5th ed., Ch. 3 (Sec. 3.1–3.5) and Ch. 4 (discrete-time stochastic processes)
 - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 3–4
- Monte Carlo methods in finance: [Glasserman \(2003\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_04](#)
 - ▶ ported from the SFE Quantlets of the textbook (SFErandu, SFErangen1, SFErangen2, SFEBinomp, SFEWienerProcess, SFEsimGBM)
 - ▶ each chart in these slides links to the Quantlet that draws it
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Statistics of Financial Markets](#)

1654: the Problem of Points

- Two players stop a fair game before the end: how should the stake be split?
- Blaise Pascal and Pierre de Fermat solved it in letters (1654): split by the probability of winning, not by the score
- Example: A needs 1 more win, B needs 2; at most two more rounds
 - ▶ B wins only if B wins both rounds: probability $1/4$
 - ▶ A should get 75% of the stake: its **expected** gain
- The same idea prices a derivative today: a fair price is an expected payoff

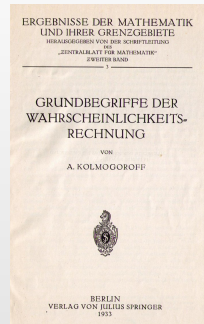


Blaise Pascal (1623–1662)

Portrait: unknown author, after François II Quesnel (c. 1690); public domain; Wikimedia Commons

Kolmogorov's Axioms (1933)

- A **probability space** $(\Omega, \mathcal{F}, P)$ Kolmogoroff (1933)
 - ▶ Ω : all possible outcomes (all possible paths of tomorrow's market)
 - ▶ $\mathcal{F}$: the events we can assign a probability to (a σ -algebra)
 - ▶ P : a function from events to $[0, 1]$
- Axioms
 - ▶ $P(A) \geq 0$; $P(\Omega) = 1$
 - ▶ for disjoint $A_1, A_2, \dots$: $P(\cup_i A_i) = \sum_i P(A_i)$
- Everything else (expectation, independence, processes) is built on these three rules



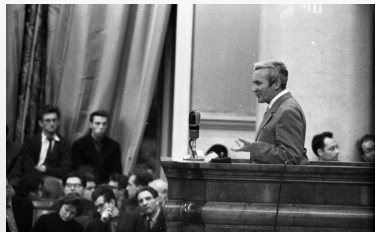
Title page of the *Grundbegriffe*, Springer 1933

Photo: KurtSchwitters (2012); CC BY-SA 3.0;

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Andrey Kolmogorov

- ▣ Andrey Nikolaevich Kolmogorov (1903–1987), Moscow State University
- ▣ Made probability a branch of measure theory: events are sets, probability is a measure
- ▣ Also: conditional expectation in its modern form, Markov processes, turbulence, complexity
- ▣ The Kolmogorov–Smirnov test ([Chapter 6](#)) carries his name



Kolmogorov lecturing in Moscow, 1963–1964

Photo: Vsevolod Tarasevich (1963–1964); CC BY 4.0; Wikimedia Commons

Events, Conditional Probability, Independence

- ▣ **Conditional probability:** $P(A | B) = P(A \cap B)/P(B)$, for $P(B) > 0$
- ▣ **Bayes' rule:** $P(B | A) = P(A | B)P(B)/P(A)$
 - ▶ reverses the direction of conditioning: from “signal given crisis” to “crisis given signal”
- ▣ **Independent events:** $P(A \cap B) = P(A)P(B)$, i.e. $P(A | B) = P(A)$
- ▣ S&P 500, 2000–2026: $P(\text{down day}) = 46.3\%$; $P(\text{down} | \text{down yesterday}) = 43.4\%$
 - ▶ two down days in a row: 20.1%, against 21.4% under independence
 - ▶ BET: 50.0% after a down day vs 42.6% after an up day: thin trading makes signs persist

Recap: From Games to Axioms

- ▣ Probability space $(\Omega, \mathcal{F}, P)$ and three axioms
- ▣ Fair value = expected value: the idea of Pascal and Fermat
- ▣ Conditional probability measures how information changes the odds

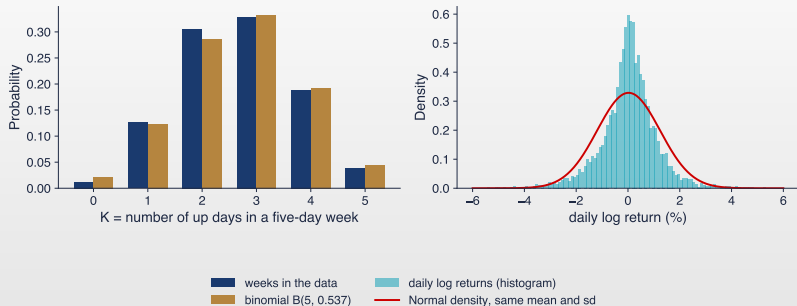
Random Variables

- A **random variable** $X : \Omega \rightarrow \mathbb{R}$ assigns a number to each outcome
 - ▶ tomorrow's log return, the number of up days in a week, the loss of a portfolio
- **Discrete**: finitely or countably many values x_i
 - ▶ **PMF** (probability mass function): $p(x_i) = P(X = x_i)$, with $\sum_i p(x_i) = 1$
- **Continuous**: $P(a < X \leq b) = \int_a^b f(x) dx$
 - ▶ f is the **PDF** (probability density function), $f \geq 0$, $\int f = 1$; $P(X = x) = 0$ for every single x
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Sec. 3.1

The Distribution Function and the Quantile Function

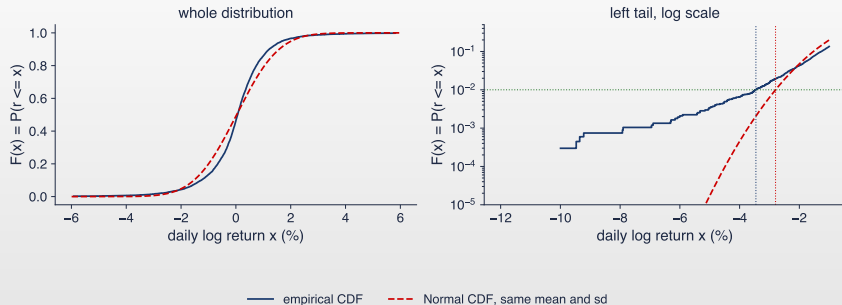
- **CDF** (cumulative distribution function): $F(x) = P(X \leq x)$
 - ▶ non-decreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$; $f = F'$ for a continuous X
- **Quantile** of order α : $q_\alpha = F^{-1}(\alpha) = \inf\{x : F(x) \geq \alpha\}$
 - ▶ median = $q_{0.5}$; the 1% quantile is the return beaten 99% of the days
- **VaR** (Value at Risk) 1%: $\text{VaR}_{1\%} = -q_{0.01}$, the loss exceeded with probability 1%
 - ▶ a quantile of the return distribution; estimation and backtesting in [Chapter 10](#)
- The **empirical CDF** of a sample: $\hat{F}(x) = \frac{1}{n} \#\{t : r_t \leq x\}$

A Discrete and a Continuous Random Variable



- Left: $K = \text{up days in a full week of the S\&P 500 (1 144 weeks)}$ vs $B(5, p)$ with $p = 0.537$: mean 2.67 vs 2.69, variance 1.16 vs 1.24
- Right: daily log returns, a continuous random variable; the histogram estimates the PDF
- The binomial fits K well; the Normal density misses the peak and the tails of the returns ([Chapter 2](#))

Empirical CDF and the 1% Quantile of the S&P 500



- $n = 6714$ days; median 0.063%; $\hat{F}(0) = 46.3\%$ of the days are not up days
- VaR 1%: empirical 3.45%, Normal 2.80%; $P(r < -3\%)$: 1.55% vs 0.63%
- $P(r < -5\%)$: 0.34% in the data, 199 times the value under the Normal distribution ($1.7 \times 10^{-3}\%$)

Recap: Random Variables

- PMF for discrete, PDF for continuous variables; the CDF works for both
- Quantiles invert the CDF; VaR 1% is minus the 1% quantile
- The tails of the empirical CDF are much heavier than the Normal ones

Expectation

- **Expectation** (mean): $E[X] = \sum_i x_i p(x_i)$ or $E[X] = \int x f(x) dx$
 - ▶ the centre of mass of the distribution; it exists only if $E|X| < \infty$ (not for Cauchy, [Chapter 3](#))
- Of a function: $E[g(X)] = \int g(x) f(x) dx$, without finding the distribution of $g(X)$
- **Linearity**: $E[aX + bY + c] = aE[X] + bE[Y] + c$, always (no independence needed)
 - ▶ the expected return of a portfolio is the weighted mean of the expected returns
- In general $E[g(X)] \neq g(E[X])$: $E[e^X] > e^{E[X]}$ (Jensen), the volatility drag of [Chapter 1](#)

Variance and Moments

- **Variance:** $\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$; standard deviation $\sigma = \sqrt{\text{Var}(X)}$ (volatility)
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$: a shift does not change risk, a scale does
- Moments $E[X^k]$; skewness and kurtosis use $k = 3, 4$ ([Chapter 2](#))
- Worked example: S&P 500 daily mean 0.0247%, s.d. 1.213%, 251 days a year
 - ▶ for i.i.d. daily returns (independent and identically distributed): $E[\sum r_t] = 251 \times 0.0247 = 6.2\%$ a year
 - ▶ $\text{Var}(\sum r_t) = 251 \sigma^2$, so annual volatility $1.213\sqrt{251} = 19.2\%$: the square-root-of-time rule

How Rare Is a 4-Sigma Day? Chebyshev's Bound

- ▣ **Chebyshev's inequality:** for any X with finite variance, $P(|X - \mu| \geq k\sigma) \leq 1/k^2$
 - ▶ it uses only the mean and the variance, so it holds for every distribution
- ▣ $k = 4$, S&P 500, 2000–2026
 - ▶ Chebyshev bound: at most 6.25% of the days
 - ▶ Normal distribution: 0.0063%
 - ▶ data: 46 days, 0.69%
- ▣ Reality is between the two: the Normal distribution is far too optimistic, the worst case of Chebyshev too pessimistic

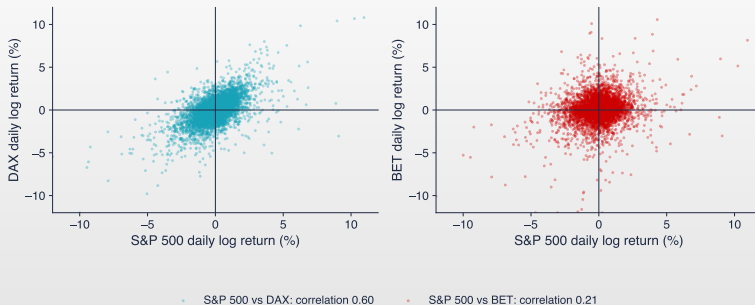
Recap: Expectation and Variance

- ▣ Expectation is linear; variance scales with the square
- ▣ For i.i.d. returns the mean grows with h and the volatility with $\sqrt{h}$
- ▣ Chebyshev bounds tail probabilities for any distribution

Joint Distribution, Covariance, Correlation

- **Joint CDF:** $F(x, y) = P(X \leq x, Y \leq y)$; marginals $F_X(x) = F(x, \infty)$
 - ▶ joint PDF $f(x, y)$; marginal $f_X(x) = \int f(x, y) dy$
- **Covariance:** $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$
- **Correlation:** $\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y) \in [-1, 1]$
 - ▶ measures **linear** dependence only; $|\rho| = 1$ iff $Y = a + bX$
- **Variance of a sum:** $\text{Var}(aX + bY) = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \text{Cov}(X, Y)$
 - ▶ with weights w and covariance matrix Σ : $\sigma_p^2 = w^\top \Sigma w$

Joint Distributions of Daily Returns



- Common trading days only: prices are joined first, then differenced ($n = 6\,605$ days for S&P 500 and DAX)
- S&P 500 and DAX: $\rho = 0.60$; S&P 500 and BET: $\rho = 0.21$
- The cloud is elliptic in the centre, but extreme days are joint: crises hit all markets together

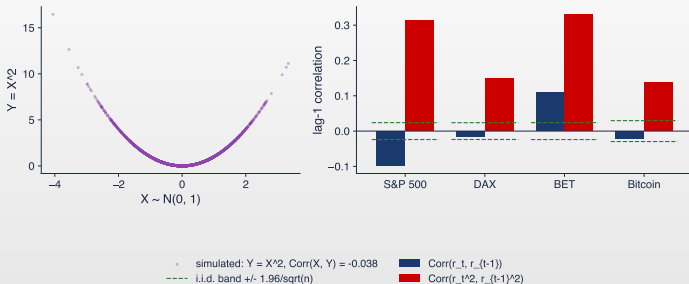
Worked Example: a 50/50 Portfolio of S&P 500 and DAX

- Daily s.d.: $\sigma_1 = 1.22\%$, $\sigma_2 = 1.43\%$; covariance 1.045; correlation 0.60
- $\sigma_p^2 = 0.25\sigma_1^2 + 0.25\sigma_2^2 + 2 \times 0.25 \text{Cov} = 0.883 + 0.522 = 1.405$
 - ▶ $\sigma_p = 1.19\%$, below the average volatility 1.32%: diversification
- S&P 500 and BET, lower correlation: $\sigma_p = 1.04\%$ vs average 1.33%, a larger gain
- Diversification works only through $\rho < 1$; in a crash correlations rise, exactly when it is needed

Independence and Uncorrelatedness

- X and Y are **independent** if $F(x, y) = F_X(x)F_Y(y)$ for all x, y
 - ▶ then $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ for **all** functions g, h
- Independent $\Rightarrow$ uncorrelated ($g, h = \text{identity}$); the converse is false
- Example: $X \sim N(0, 1)$, $Y = X^2$
 - ▶ $\text{Cov}(X, X^2) = E[X^3] = 0$: uncorrelated
 - ▶ but Y is a function of X : completely dependent
- Exception: for jointly Normal variables, uncorrelated $\Leftrightarrow$ independent

Returns Are Uncorrelated, Squared Returns Are Not



- Left: simulated $Y = X^2$, sample correlation -0.038 , yet Y is determined by X
- Right: $\text{Corr}(r_t, r_{t-1})$ is small (S&P 500 -0.099 , DAX -0.017 , Bitcoin -0.022); $\text{Corr}(r_t^2, r_{t-1}^2)$ is far outside the band: 0.31, 0.15, 0.14
- BET: $\text{Corr}(r_t, r_{t-1}) = 0.109$, a trace of thin trading; the negative S&P 500 value comes mostly from crisis days

Recap: Joint Distributions

- ▣ Covariance enters the variance of every portfolio
- ▣ Independent implies uncorrelated, not the other way round
- ▣ Returns: nearly uncorrelated, but their squares are correlated, so they are not independent

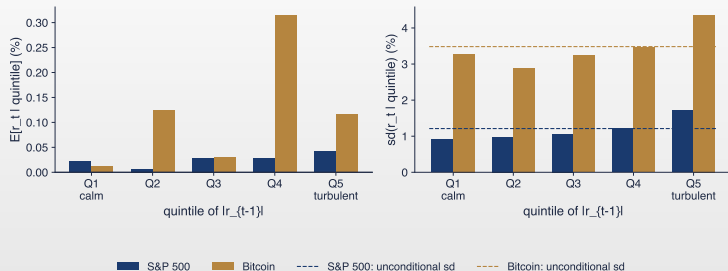
Conditional Expectation

- Conditional density: $f(y | x) = f(x, y)/f_X(x)$
 - ▶ the distribution of Y once we know that $X = x$
- **Conditional expectation:** $E[Y | X = x] = \int y f(y | x) dy$; $E[Y | X]$ is a random variable, a function of X
- It is the best forecast of Y from X : it minimises $E[(Y - g(X))^2]$ over all functions g
 - ▶ for returns: $E[r_t | \mathcal{F}_{t-1}]$, the forecast given all information $\mathcal{F}_{t-1}$ up to yesterday
- **Tower property** (law of iterated expectations): $E[E[Y | X]] = E[Y]$

Conditional Variance and the Law of Total Variance

- **Conditional variance:** $\text{Var}(Y | X) = E[(Y - E[Y | X])^2 | X]$
- **Law of total variance:** $\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$
 - ▶ within-group variance plus between-group variance
- Worked example, two regimes: calm (probability 0.7, mean 0.05%, s.d. 0.8%), turbulent (0.3, -0.10%, 2%)
 - ▶ $E[r] = 0.7 \times 0.05 + 0.3 \times (-0.10) = 0.005\%$
 - ▶ $E[\text{Var}] = 0.7 \times 0.64 + 0.3 \times 4 = 1.648$; $\text{Var}(E) = 0.0047$
 - ▶ $\text{Var}(r) = 1.653$, s.d. 1.286%: almost all risk comes from the changing variance, not from the changing mean

Today's Return Given Yesterday's Move



- ▣ Quintiles of $|r_{t-1}|$: Q1 = the calmest 20% of yesterdays, Q5 = the most turbulent 20%
- ▣ Conditional mean: between 0.006% and 0.042% for the S&P 500, no usable pattern
- ▣ Conditional s.d.: S&P 500 from 0.91% (Q1) to 1.71% (Q5), ratio 1.88; Bitcoin ratio 1.33
- ▣ Law of total variance, S&P 500: $1.469 = 1.469 + 0.00013$; the between part is 0.01%

From Conditional Variance to GARCH

- Write the return as $r_t = \mu_t + \sigma_t z_t$, with z_t i.i.d., mean 0, variance 1
 - ▶ $\mu_t = E[r_t | \mathcal{F}_{t-1}]$: the conditional mean, close to constant
 - ▶ $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$: the conditional variance, which changes with the news
- ARCH (autoregressive conditional heteroskedasticity) Engle (1982): $\sigma_t^2 = \omega + \alpha r_{t-1}^2$
 - ▶ GARCH Bollerslev (1986): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ (Chapter 9)
- Exactly the pattern in the chart: a large $|r_{t-1}|$ raises today's conditional s.d.
- The unconditional distribution is a mixture over σ_t : heavier tails than the Normal distribution, even if z_t is Normal

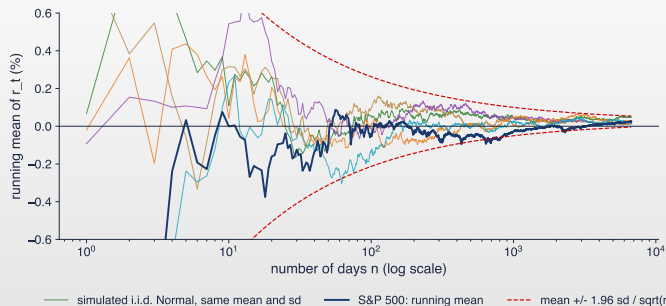
Recap: Conditioning

- ▣ $E[Y | X]$ is the best forecast; the tower property averages it back to $E[Y]$
- ▣ Returns: conditional mean almost flat, conditional variance strongly time-varying
- ▣ Modelling σ_t^2 is the job of ARCH and GARCH

Two Limit Theorems

- ▣ **LLN** (law of large numbers): for i.i.d. X_i with $E|X| < \infty$, $\bar{X}_n \rightarrow \mu$ as $n \rightarrow \infty$
 - ▶ the reason why averages, frequencies and Monte Carlo estimates work
- ▣ **CLT**: with finite variance, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \rightarrow N(0, 1)$ ([Chapter 2](#))
 - ▶ gives the speed: the error of $\bar{X}_n$ is about $\sigma/\sqrt{n}$
 - ▶ infinite variance: sums converge to α -stable laws instead ([Chapter 3](#))
- ▣ Both need some form of independence; dependent returns converge more slowly

The LLN for the Mean Return: Slow



- S&P 500, 26.7 years: mean 0.0247% a day, standard error $\sigma / \sqrt{n} = 0.0148\%$, $t = 1.67$
- Annual mean 6.2%, 95% interval $[-1.1\%, 13.5\%]$: even 26 years do not pin down the expected return
- For $t = 3$ at this mean and volatility we would need about 86 years of data

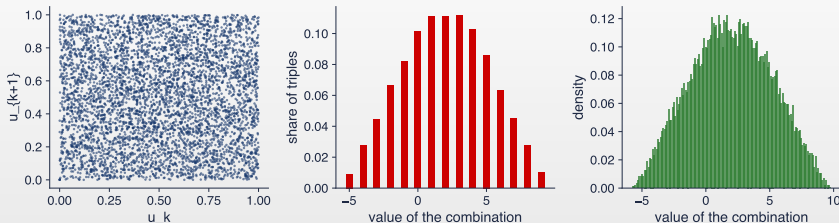
The Monte Carlo Idea

- Estimate $\theta = E[g(X)]$ by $\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^N g(X_i)$ with simulated X_i
 - ▶ a probability is an expectation: $P(A) = E[1_A]$
 - ▶ the LLN makes $\hat{\theta}_N \rightarrow \theta$; the CLT gives the error $\text{sd}(g(X))/\sqrt{N}$
- Named after the Monaco casino by Ulam and von Neumann, Los Alamos, 1940s [Metropolis & Ulam \(1949\)](#)
- In finance [Glasserman \(2003\)](#)
 - ▶ prices of path-dependent options, VaR and ES of portfolios, stress scenarios
 - ▶ any expectation without a closed form
- Two ingredients: uniform random numbers and a way to turn them into any distribution

Pseudo-Random Numbers: Linear Congruential Generators

- **LCG** (linear congruential generator): $x_{k+1} = (ax_k + c) \bmod M$, $u_k = x_k/M \in [0, 1)$
 - ▶ deterministic: the same seed x_0 gives the same sequence, which makes results reproducible
- Worked example (SFERangen1): $a = 5$, $c = 1$, $M = 16$, $x_0 = 7$
 - ▶ x : 7, 4, 5, 10, 3, 0, 1, 6, 15, 12, 13, 2, 11, 8, 9, 14, 7
 - ▶ period 16, the maximum M ; with $a = 4$ the sequence sticks after one step (period 1)
- Real generators use huge M ; the period must be far longer than any simulation
- Good uniform marginals are not enough: consecutive numbers must also look independent

RANDU: Random Numbers Fall Mainly in the Planes (SFERandu)



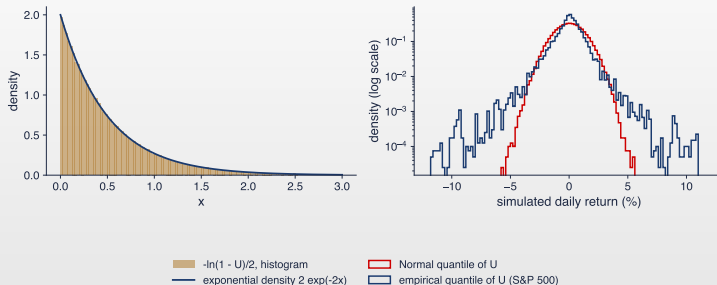
• RANDU pairs (u_k, u_{k+1}) ■ RANDU: $9u_k - 6u_{k+1} + u_{k+2}$ ■ PCG64: the same combination

- RANDU, IBM, 1960s: $x_{k+1} = 65539 x_k \bmod 2^{31}$; pairs look uniform (left), correlation 0.003
- But $9u_k - 6u_{k+1} + u_{k+2}$ takes only 15 integer values, from -5 to 9: all triples lie on 15 planes [Marsaglia \(1968\)](#)
- Modern generators: Mersenne Twister [Matsumoto & Nishimura \(1998\)](#), PCG64 (the NumPy default, right panel)

The Inverse Transform Method

- **Theorem:** if $U \sim U(0, 1)$ and F is a CDF, then $X = F^{-1}(U)$ has CDF F
 - ▶ proof: $P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$
- Exponential with rate λ : $F(x) = 1 - e^{-\lambda x}$, so $X = -\ln(1 - U)/\lambda$
 - ▶ $\lambda = 2$, $u = 0.9$: $x = -\ln(0.1)/2 = 1.151$
- Normal: $X = \mu + \sigma \Phi^{-1}(U)$
- Data: use the empirical quantile function $\hat{F}^{-1}$
 - ▶ this is resampling past days: historical simulation and the bootstrap

Inverse Transform: Exponential, Normal and Historical

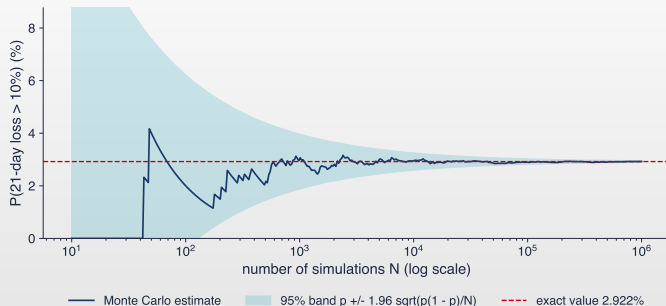


- Left: 200 000 draws — $-\ln(1 - U)/2$ match the exponential density; sample mean 0.500 (theory $1/\lambda = 0.5$)
- Right: the same uniforms through the Normal and the empirical S&P 500 quantile functions
- Draws below -5% : 0.0035% Normal vs 0.34% historical (data: 0.34%); worst draw -5.7% vs -12.7%

Worked Example: a Monthly Loss Probability by Monte Carlo

- Model: S&P 500 daily log returns i.i.d. $N(0.0247, 1.213^2)$ (in %); event: 21-day log return below -10%
- Exact: the sum is $N(0.519, 5.56^2)$, so $p = \Phi(-1.892) = 2.922\%$
- Monte Carlo: $\hat{p}_N$ = share of simulated months below -10% ; standard error $\sqrt{p(1-p)/N}$
 - ▶ $N = 1\,000$: 3.000% (s.e. 0.533%); $N = 100\,000$: 2.901% (s.e. 0.053%)
 - ▶ 10 times more precision needs 100 times more simulations
- For a relative error of 10% (95% level) we need $N \approx 1.96^2(1-p)/(0.1^2p) = 12\,762$

Monte Carlo Converges at Rate $1/\sqrt{N}$



- The estimate wanders for small N and settles inside the 95% band, which shrinks like $1/\sqrt{N}$
- $N = 10^6$: 2.926% vs exact 2.922%, s.e. 0.017%
- Monte Carlo removes the simulation error, not the model error: Seminar 4 (B3) repeats it with resampled days

Recap: Random Numbers and Monte Carlo

- ▣ Pseudo-random generators are deterministic; check them in several dimensions
- ▣ $F^{-1}(U)$ turns uniforms into any distribution, including the empirical one
- ▣ Monte Carlo error $\propto 1/\sqrt{N}$; always report it

The Binomial Process

- Each period the price moves up by a factor u with probability p or down by $d < u$ with probability $1 - p$, independently
 - ▶ after n steps with K up moves: $S_n = S_0 u^K d^{n-K}$, $K \sim B(n, p)$
- $\ln S_n = \ln S_0 + K \ln u + (n - K) \ln d$: a random walk in log prices (Section 9)
- $E[S_n] = S_0 (pu + (1 - p)d)^n$, because the steps are independent
 - ▶ the basis of the binomial option pricing model [Cox, Ross & Rubinstein \(1979\)](#); textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 7

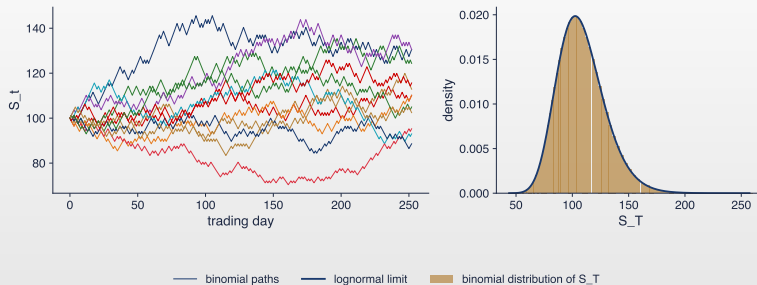
Worked Example: a Two-Step Tree

- $S_0 = 100$, $u = 1.2$, $d = 0.8$, $p = 0.6$
- Terminal values and probabilities
 - ▶ uu : 144 with $p^2 = 0.36$; ud or du : 96 with $2p(1-p) = 0.48$; dd : 64 with $(1-p)^2 = 0.16$
- $E[S_2] = 100 \times (0.6 \times 1.2 + 0.4 \times 0.8)^2 = 100 \times 1.04^2 = 108.16$
- $P(S_2 < 100) = 0.48 + 0.16$: an up and a down move lose money, because $ud = 0.96 < 1$
- With $p^* = (1-d)/(u-d) = 0.50$ the price is a fair game, $E[S_{t+1} | S_t] = S_t$ (Section 9)

Calibrating the Tree to a Market (Cox–Ross–Rubinstein)

- **CRR** (Cox–Ross–Rubinstein) with step Δt : $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $p = (e^{\mu\Delta t} - d)/(u - d)$
 - ▶ so that the mean and the variance of each step match a market with drift μ and volatility σ
- S&P 500, 2000–2026: $\sigma = 19.23\%$, $\mu = 8.06\%$ a year; daily steps, $\Delta t = 1/252$
 - ▶ $u = 1.0122$, $d = 0.9880$, $p = 0.5102$
- As $\Delta t \rightarrow 0$, $\ln S_T$ becomes Normal (CLT for K): the tree converges to GBM (Section 10)

Binomial Paths and the Lognormal Limit (SFEBinomp)



- Left: 12 paths over one year of daily steps, $S_0 = 100$
- Right: distribution of S_T with 252 steps; $E[S_T] = 108.40 = 100e^\mu$; 5% quantile 78.5 vs lognormal 77.6
- $P(S_T < 100)$: 39.7% binomial vs 37.3% lognormal ($S_T = 100$ exactly has a probability mass, which the binomial counts on one side)

Recap: The Binomial Model

- Up or down each period: $S_n = S_0 u^K d^{n-K}$ with binomial K
- CRR matches μ and σ ; the limit is lognormal
- A special p^* turns the price into a fair game: the key to option pricing

Stochastic Processes and Information

- A **stochastic process** $\{X_t\}$: a family of random variables indexed by time
 - ▶ one realisation = one path; a price series is a single path of the process
- $\mathcal{F}_t$: the information available at time t (past prices, news); it grows with t (a **filtration**)
- **(Weakly) stationary**: $E[X_t]$ and $\text{Cov}(X_t, X_{t+h})$ do not depend on t
 - ▶ returns: roughly yes; prices: no
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 4 and Sec. 11.3

White Noise and the Random Walk

- **White noise:** $E[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$, $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$
 - ▶ i.i.d. noise is stronger: the ε_t are independent, not only uncorrelated
- **Random walk:** $S_t = S_{t-1} + \mu + \varepsilon_t = S_0 + \mu t + \sum_{s=1}^t \varepsilon_s$
 - ▶ $E[S_t] = S_0 + \mu t$; $\text{Var}(S_t) = t\sigma^2$: not stationary
 - ▶ simulated with $\sigma = 1$: variance 10.1, 49.8, 100.9 at $t = 10, 50, 100$
- Log prices as a random walk = returns as white noise: the benchmark of [Chapter 7](#) (efficient markets)

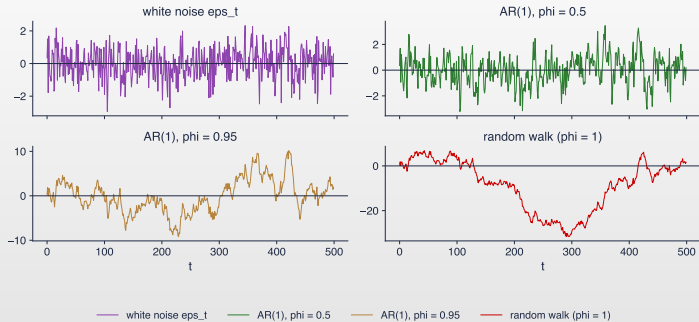
Martingales

- $\{X_t\}$ is a **martingale** if $E|X_t| < \infty$ and $E[X_{t+1} | \mathcal{F}_t] = X_t$
 - ▶ a fair game: the best forecast of tomorrow is today
- A **martingale difference**: $\varepsilon_t = X_t - X_{t-1}$ with $E[\varepsilon_t | \mathcal{F}_{t-1}] = 0$
- Examples
 - ▶ a random walk without drift; $S_t^2 - t\sigma^2$ for the same walk; the binomial price under p^*
- Martingale $\neq$ i.i.d. increments: the conditional **variance** may change, as in GARCH
 - ▶ the efficient-market idea in its weak form [Fama \(1970\)](#): excess returns are a martingale difference

The AR(1) Process

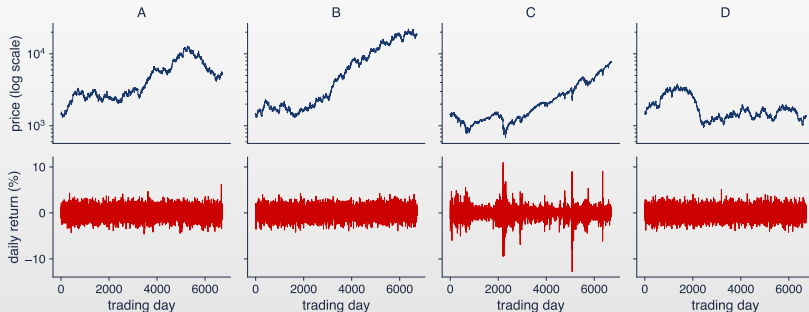
- **AR(1)** (autoregressive of order 1): $X_t = c + \phi X_{t-1} + \varepsilon_t$, ε_t white noise
 - ▶ $\phi = 0$: white noise; $\phi = 1$, $c = \mu$: random walk
- Stationary iff $|\phi| < 1$; then
 - ▶ mean $c/(1 - \phi)$, variance $\sigma^2/(1 - \phi^2)$, autocorrelation $\rho(h) = \phi^h$
 - ▶ half-life of a shock: $\ln 0.5 / \ln \phi$ periods
- Simulated, $\sigma = 1$
 - ▶ $\phi = 0.5$: variance 1.34 (theory 1.33); $\rho(5) = 0.034$ (theory 0.031); half-life 1.0
 - ▶ $\phi = 0.95$: variance 10.20 (theory 10.26); $\rho(5) = 0.772$; half-life 13.5
- Uses: interest rates and spreads (mean reversion); the squared returns behave like a persistent AR, which leads to GARCH

White Noise, AR(1) and Random Walk, Same Shocks



- The larger ϕ , the longer a shock lives: the AR(1) with $\phi = 0.95$ wanders, but returns to 0
- The random walk never returns on purpose: shocks are permanent and its variance grows with t

Question for the Room: Which One Is Real?



- One column is the S&P 500, 2000–2026; three are GBM paths (i.i.d. Normal returns) with the same mean and volatility
- **What do you think?** Which column is real, from the prices alone?
- **What do you think?** Which column is real, from the returns?

Answer: Column C

- **Answer:** column C is the S&P 500
- Prices: hard to tell; random walks with drift produce booms and long falls too
 - ▶ this is why [Bachelier \(1900\)](#) and [Osborne \(1959\)](#) modelled prices as random walks
- Returns: obvious; the real series has quiet years and violent clusters (2008, 2020)
 - ▶ excess kurtosis 10.7 vs at most 0.02 in the simulations; $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31$ vs at most 0.006
 - ▶ largest absolute daily move 12.8% vs 6.2%
- Lesson: a model can fit the price path and still be wrong about risk

Recap: Discrete-Time Processes

- White noise: uncorrelated; i.i.d. noise: independent; a martingale difference: unpredictable mean
- Random walk: permanent shocks, variance $t\sigma^2$; AR(1) with $|\phi| < 1$: shocks fade at rate ϕ^h
- Prices look like random walks; returns are not i.i.d.

From Pollen to Prices: Brownian Motion

- 1827: the botanist Robert Brown sees pollen particles jiggling in water
- 1900: Louis Bachelier models the Paris bond market with the same motion, five years before physics [Bachelier \(1900\)](#)
- 1905: Albert Einstein explains it by molecular collisions [Einstein \(1905\)](#)
- 1923: Norbert Wiener builds the rigorous mathematical object [Wiener \(1923\)](#)
- 1959: [Osborne \(1959\)](#) applies it to log prices: geometric Brownian motion



Robert Brown (1773–1858)

Photo: [Mauil & Polyblank \(1855\)](#); public domain;
[Wikimedia Commons](#)

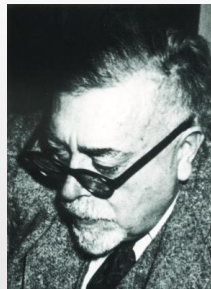


Louis Bachelier (1870–1946)

Photo: [unknown author](#); public domain; [Wikimedia Commons](#)

The Wiener Process

- $\{W(t), t \geq 0\}$ is a **Wiener process** (standard Brownian motion) if
 - ▶ $W(0) = 0$
 - ▶ increments over disjoint intervals are independent
 - ▶ $W(t) - W(s) \sim N(0, t - s)$ for $s < t$
 - ▶ the paths are continuous
- It is the limit of the scaled random walk $W_n(t) = (\varepsilon_1 + \dots + \varepsilon_{\lfloor nt \rfloor}) / \sqrt{n}$, $\varepsilon = \pm 1$ (Donsker)
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 5

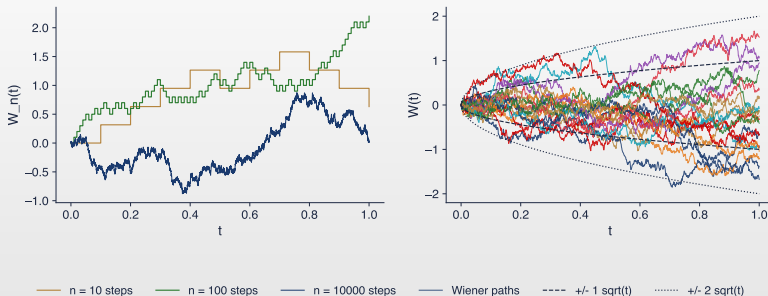


Norbert Wiener (1894–1964)

Photo: [Konrad Jacobs](#); CC BY-SA 2.0 DE;

[Wikimedia Commons](#)

From Random Walks to the Wiener Process (SFEWienerProcess)



- Left: scaled random walks with 10, 100 and 10 000 steps; the steps vanish, the randomness stays
- Right: 20 Wiener paths; $\text{Var}(W(t)) = t$, so they spread like $\pm \sqrt{t}$; simulated $\text{Var}(W(1)) = 1.006$; 68.2% of $W(1)$ inside ± 1 (theory 68.3%)

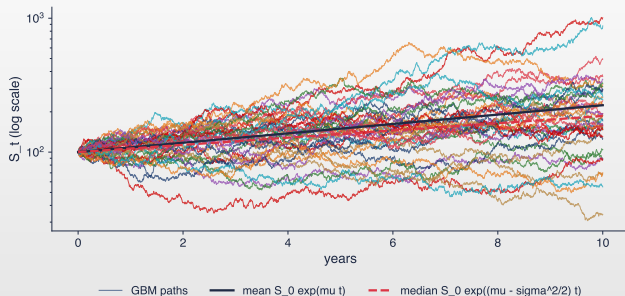
Strange Properties of Wiener Paths

- Continuous everywhere, differentiable nowhere
 - ▶ $(W(t+h) - W(t))/h \sim N(0, 1/h)$ explodes as $h \rightarrow 0$
- **Quadratic variation:** $\sum_k (W(t_{k+1}) - W(t_k))^2 \rightarrow t$ on a finer and finer grid
 - ▶ 1000 steps on $[0, 1]$: mean 0.999, s.d. 0.045 across 10 000 paths
 - ▶ the reason why $(dW)^2 = dt$ in Itô calculus, and why realised variance measures volatility ([Chapter 8](#))
- $\text{Cov}(W(s), W(t)) = \min(s, t)$: simulated $\text{Cov}(W(0.5), W(1)) = 0.496$
- W is a martingale; so is $W(t)^2 - t$

Geometric Brownian Motion

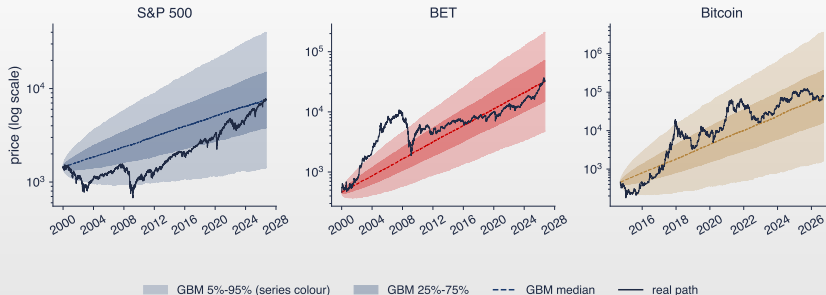
- **GBM** (geometric Brownian motion): $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ▶ the relative change dS/S has drift μ and volatility σ
- Solution: $S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t)$
 - ▶ $\ln(S_t/S_0) \sim N((\mu - \sigma^2/2)t, \sigma^2 t)$: lognormal prices, i.i.d. Normal log returns
- Mean $E[S_t] = S_0 e^{\mu t}$; median $S_0 e^{(\mu - \sigma^2/2)t}$: the gap is the volatility drag
- Exact simulation on a grid: $S_{t+\Delta t} = S_t \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z)$, $Z \sim N(0, 1)$
- The model behind [Black & Scholes \(1973\)](#) ([Chapter 6](#) of the textbook)

GBM Calibrated to the S&P 500 (SFEsimGBM)



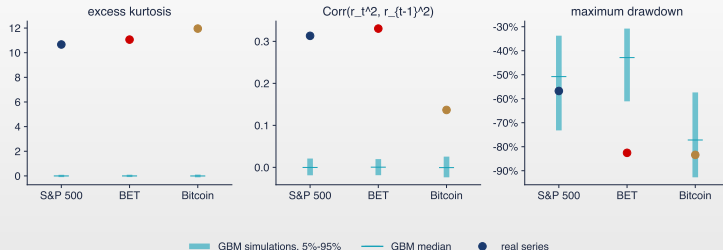
- Calibration, 2000–2026: $\sigma = 19.23\%$, log drift $\mu - \sigma^2/2 = 6.21\%$, so $\mu = 8.06\%$ a year
- After 10 years: mean 224.0, median 186.2; 62.0% of the paths end below the mean (theory $\Phi(\sigma\sqrt{T}/2) = 61.9\%$)
- Probability of a loss after 10 years: 15.3%

Real Paths in the GBM Fan



- Each fan: 4000 GBM paths with the drift and volatility of the series itself, started on its first day
- S&P 500: the path reached the 0.5% quantile on 23 July 2002, then the median at the end; outside the 90% band 7% of the time
- BET: outside the band 24% of the time; the 2000–2007 boom and the 2008 crash are not GBM-like; Bitcoin: $\sigma = 66.6\%$ a year makes the fan enormous

What GBM Misses

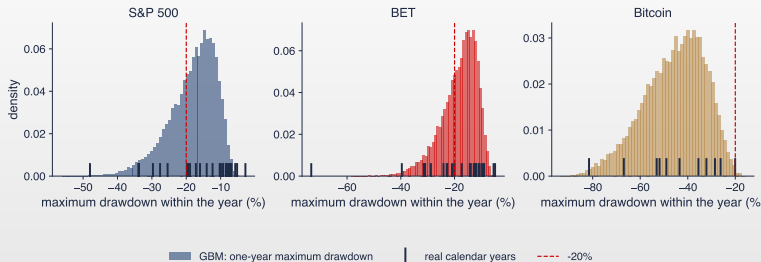


- Excess kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0; 90% of 300 GBM simulations lie within $[-0.10, 0.09]$
- Volatility clustering: $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31, 0.33, 0.14$ vs at most 0.03 under GBM
- Maximum drawdown: S&P 500 -57% is ordinary for GBM; BET -83% is beyond all but 5% of the paths (5% quantile -61%)
- 4-sigma days: 46 (S&P 500) vs at most 1 in 95% of the GBM paths; crypto has the same problem [Pele et al. \(2023\)](#)

Recap: Wiener Process and GBM

- Wiener process: independent Normal increments with variance equal to the elapsed time
- GBM: lognormal prices, median below mean, the benchmark of option pricing
- Real returns have heavy tails and volatility clusters that GBM cannot produce

An Open Question: Does GBM Get Drawdown Risk Right?



- Drawdown within a calendar year beyond 20%: S&P 500 in 6 of 26 years (23%) vs 34% under GBM; BET 46% vs 34%
- Beyond 40%: GBM expects 0.14 such years for the S&P 500 and 0.17 for the BET; both had 1 (2008: -48% and -73%)
- Open: does GBM overstate moderate drawdowns and understate extreme ones because calm and crisis years cluster?

How AI Could Help

- **Literature:** find studies of drawdown distributions and of their link to volatility clustering, and summarise them
- **Code:** draft simulations of yearly drawdowns under GBM, the i.i.d. bootstrap and a GARCH model
- **Design:** propose a test that compares the observed number of bad years with the model probability
- **Example prompt**
 - ▶ Write Python code that simulates 10,000 one-year paths of daily S&P 500 returns under GBM and under an i.i.d. bootstrap of 2000-2026 returns, and returns the probability of a drawdown beyond 20% and 40% within the year.

What to Check

- ▣ Definitions: drawdown from the running peak within the year, not from the first day of the year
- ▣ Calibration: the same μ and σ , the same number of trading days a year as the data
- ▣ Inference: 26 years is a small sample; compute a binomial p -value, not only shares
- ▣ Mechanism: the i.i.d. bootstrap keeps heavy tails but removes clustering; the difference isolates the effect of clustering
- ▣ References: every cited paper must exist; check the DOI

Project Seed

- **Question:** how wrong is GBM about the probability of a bad year, and why?
 - ▶ data: S&P 500, DAX, BET and Bitcoin from the course data
- Steps
 - ▶ compute within-year maximum drawdowns for every calendar year
 - ▶ simulate the same statistic under GBM, the i.i.d. bootstrap and a GARCH(1,1) model ([Chapter 9](#))
 - ▶ compare the shares beyond 20% and 40% with binomial tests
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected [Wang et al. \(2023\)](#)

Key Takeaways

- ▣ Distributions: PMF, PDF, CDF and quantiles; VaR 1% is minus a quantile
- ▣ Covariance drives portfolio risk; uncorrelated returns are not independent
- ▣ Conditional variance changes with yesterday's news: the starting point of GARCH
- ▣ Monte Carlo: inverse transform plus the LLN; error $\propto 1/\sqrt{N}$
- ▣ Binomial tree $\rightarrow$ GBM; random walk, martingale, white noise, AR(1)
- ▣ GBM fits price paths roughly, but misses heavy tails, clustering and extreme drawdowns

Key Formulas

Quantity	Formula
Variance of a sum	$\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\text{Cov}(X, Y)$
Correlation	$\rho = \text{Cov}(X, Y)/(\sigma_X\sigma_Y)$
Tower property	$E[E[Y X]] = E[Y]$
Total variance	$\text{Var}(Y) = E[\text{Var}(Y X)] + \text{Var}(E[Y X])$
Inverse transform	$X = F^{-1}(U), U \sim U(0, 1)$
Monte Carlo error	$\sqrt{p(1-p)/N}$
Random walk	$S_t = S_{t-1} + \mu + \varepsilon_t, \text{Var}(S_t) = t\sigma^2$
AR(1)	$\text{Var}(X_t) = \sigma^2/(1 - \phi^2), \rho(h) = \phi^h$
GBM	$S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t), E[S_t] = S_0 e^{\mu t}$

Check Yourself

- **Question:** $\sigma_1 = 2\%$, $\sigma_2 = 3\%$, $\rho = 0.5$; what is the volatility of the 50/50 portfolio?
 - ▶ **Answer:** $\sqrt{0.25 \times 4 + 0.25 \times 9 + 2 \times 0.25 \times 0.5 \times 2 \times 3} = \sqrt{4.75} = 2.18\%$
- **Question:** GBM with $\mu = 8\%$, $\sigma = 20\%$, $S_0 = 100$; what are the mean and the median of S_1 ?
 - ▶ **Answer:** mean $100e^{0.08} = 108.33$, median $100e^{0.06} = 106.18$
- **Question:** AR(1) with $\phi = 0.8$ and $\sigma = 1$; what is its variance?
 - ▶ **Answer:** $1/(1 - 0.64) = 2.78$
- Next: [Chapter 5](#), heavy tails and extreme value theory

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